

2024 Year in Review

Global Internet Traffic, Technology Adoption & Security Trends

Observed through Cloudflare's global network

Data covering January – December 2024

TRAFFIC GROWTH

17% Global Traffic Growth and Starlink's 3.3x Surge

Key Insight: Global Internet traffic grew 17% YoY, with satellite connectivity (Starlink) surging 3.3x to reach underserved regions.

GLOBAL TRAFFIC GROWTH

17%

Year-over-Year Increase

vs. 2023 baseline

STARLINK TRAFFIC GROWTH

3.3x

vs. Prior Year

Satellite connectivity expansion



Context

- Trend line normalized from mid-January to remove holiday volatility
- Starlink continued expansion to previously underserved geographic regions
- Increasing reliance on Internet for communication, commerce, entertainment & transportation

ADDITIONAL METRICS

41% of global traffic from mobile devices

28% of dual-stack traffic on IPv6

33% of traffic from iOS devices

Google Leads Top 10 Services as Generative AI Rises

Key Insight: Google tops rankings for second consecutive year; Generative AI surged throughout 2024 while the Metaverse receded.

TOP 10 INTERNET SERVICES

RANK	SERVICE	CATEGORY
#1	Google	Search/Cloud
#2	Facebook	Social
#3	Apple	Hardware/Cloud
#4	TikTok	Social/Video
#5	AWS	Cloud
#6	Microsoft	Software/Cloud
#7	Instagram	Social
#8	YouTube	Video
#9	Amazon	E-commerce
#10	WhatsApp	Messaging

KEY TREND: RISE OF GENERATIVE AI



Generative AI

Meteoric rise throughout 2024

- AI services disrupting traditional service rankings
- Metaverse receded into the background
- Methodology: aggregate DNS resolver data from 1.1.1.1

SEARCH ENGINE MARKET SHARE

Google 89% of search referral traffic



Others: Yandex 3.1% • Baidu 2.7% • Bing 2.6% • DuckDuckGo 0.95%

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Key Insight: Chrome 124's default post-quantum enablement (April 2024) drove dramatic adoption surge from ~2% to 13% of TLS 1.3 traffic.

POST-QUANTUM ENCRYPTION

TLS 1.3 Traffic

2% → 13%

After Chrome 124 default (Apr 17, 2024)

HTTP/3 ADOPTION

21%

of all traffic uses HTTP/3

QUIC-based, faster connections, encryption by default

TOP API CLIENT LANGUAGE

Go ~12%

of automated API requests

Surpassed Node.js, Python (9.6%), Java (7.4%)

API CLIENT LANGUAGE BREAKDOWN



BROWSER MARKET SHARE

Chrome Dominant

Global browser market leadership continues

MOBILE OS LANDSCAPE

Android peak: 95%+ iOS share: 33%

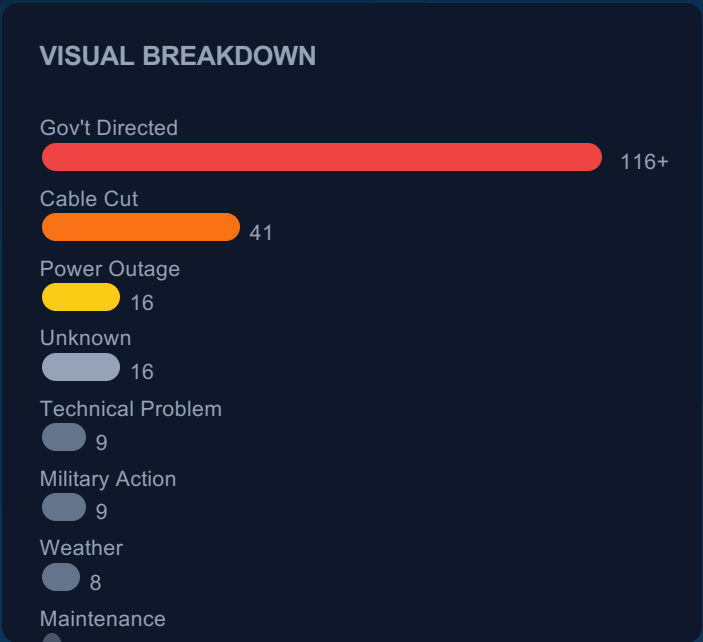
Regional variations significant; iOS peaks at ~66% in some markets

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Key Insight: Government-directed shutdowns were the #1 cause of the 225 observed Internet disruptions in 2024, reflecting increasing digital control.

225
Major Disruptions

DISRUPTIONS BY CAUSE	
CAUSE	COUNT
Government Directed	116+
Cable Cut	41
Power Outage	16
Unknown	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance / Cyber Attack	4+4



Context

- Government-directed shutdowns reflect increasing use of Internet shutdowns as a policy tool
- Cable cuts remain a significant cause (41 incidents), especially for submarine cables
- Tracked via Cloudflare Radar Outage Center using real-time traffic data

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Key Insight: 6.5% of all traffic was mitigated; Gambling/Games sector most attacked; Log4j exploitation persists 2+ years after disclosure.

TRAFFIC MITIGATED

6.5%

of all traffic processed

MOST ATTACKED SECTOR

Gambling
& Games

Industry target priority

BOT TRAFFIC CONCENTRATION

68%

from top 10 countries

MALICIOUS EMAILS

4.3%

of all emails processed

EMAIL THREAT BREAKDOWN

43% of malicious emails contained deceptive links

Top-level domain .bar originated the largest share of malicious and spam email

● CRITICAL: LOG4J PERSISTENCE

CVE-2021-44228 Log4j vulnerability still actively exploited

2+ years after December 2021 disclosure

Persistent nature of widely known vulnerabilities remains a major security challenge

RPKI VALID ROUTES GROWTH

+6.4% improving Internet routing security

TOP TLD FOR MALICIOUS EMAIL

.bar largest share of malicious/spam

GLOBAL BOT TRAFFIC

68% concentrated in just 10 countries

Five Takeaways Shaping the 2025 Internet Outlook

Summary: Key findings from 2024 that inform strategic planning for network and security leaders.

1

Traffic Growth Trajectory

17% YoY global traffic growth sustained — driven by expanding connectivity, commerce, and digital services.

2

Post-Quantum Momentum

13% adoption and climbing — Chrome 124 default driving rapid encryption u

3

Connectivity Fragility

Government shutdowns = #1 disruption cause (116+ incidents). Digital infrastructure is a policy target.

4

Persistent Security Threats

6.5% traffic mitigated; Log4j exploitation continues 2+ years post-disclosure.

5

Evolving Developer Ecosystem

Go surpasses Node.js and Python to become the #1 API client language (~12% of automated traffic).

2025 OUTLOOK INDICATORS

- ↑

Traffic growth to continue as connectivity expands
- ↑

Post-quantum encryption adoption accelerating
- ⚠

Internet shutdowns and disruptions likely to persist as policy tool
- ↑

AI services continuing meteoric rise
- ↑

Go and modern languages gaining API share
- ⚠

Legacy vulnerabilities (Log4j) remain exploitable